

Adelic S-Duality — The Kapustin-Witten Correspondence at All Places

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Abstract

Kapustin and Witten (2006) proved that S-duality in $N = 4$ super-Yang—Mills theory on a Riemann surface is equivalent to the geometric Langlands correspondence, realizing the Langlands duality at the Archimedean (∞) place. This paper extends S-duality to every p -adic place $\mathbb{Q}_p$ by constructing the **adelic S-duality transformation**. The adelic coupling $\tau_{\mathbb{A}} = (\tau_{\infty}, \tau_2, \tau_3, \dots)$ is an idele-class parameter, and S-duality acts componentwise: $\tau_v \rightarrow -1/n_{\mathfrak{g}}\tau_v$ at each place, where $n_{\mathfrak{g}}$ is the lacing number of the Lie algebra. At finite primes, S-duality becomes **anyonic level-rank duality** — the fusion ring $R(\mathrm{SU}(2))_k$ at level k maps to $R(\mathrm{SO}(3))_{\tilde{k}}$ at the dual level $\tilde{k} = k$ (for $\mathrm{su}(2)$). The global S-duality is then identified with the global Langlands correspondence for the Langlands dual group ${}^L G$ over the adèle ring $\mathbb{A}$. Four falsifiable predictions connect the adelic S-duality framework to experimental signatures.

1. Introduction

S-duality — the strong-weak coupling duality — is the most profound symmetry of four-dimensional gauge theories. In $N = 4$ super-Yang—Mills with gauge group G , S-duality maps the complexified coupling:

$$\tau = \frac{\theta}{2\pi} + \frac{4\pi i}{g^2} \longrightarrow -\frac{1}{n_{\mathfrak{g}}\tau}$$

where $n_{\mathfrak{g}} = 1$ for simply-laced Lie algebras ($\mathrm{SU}(n)$, E_6 , E_7 , E_8) and $n_{\mathfrak{g}} = 2$ for non-simply-laced ($\mathrm{SO}(2n+1)$, $\mathrm{Sp}(n)$, G_2 , F_4). [established, Kapustin—Witten 2006]

The Kapustin—Witten theorem states that S-duality with a surface operator insertion on a Riemann surface Σ is equivalent to the geometric Langlands correspondence for the Langlands dual group ${}^L G$ on Σ . This realizes geometric Langlands at the Archimedean place (spacetime $\mathbb{R}^4 \simeq \mathbb{C}^2$, Riemann surface $\Sigma \subset \mathbb{C}^2$).

Phase 1 of the Adelic Langlands Physics program established local Langlands correspondences at every non-Archimedean place p via the Silent Parameter. Phase 3.1 constructed adelic Yang-Mills theory. This paper extends S-duality to all completions of $\mathbb{Q}$.

2. Local S-Duality at Each Place

2.1 Archimedean Place ($v = \infty$)

At the ∞ -place, the adelic Yang-Mills action (Phase 3.1) reduces to the standard action on $\mathbb{R}^4$:

$$S_\infty[A] = \frac{1}{4g_\infty^2} \int_{\mathbb{R}^4} \text{Tr}(F_{\mu\nu} F^{\mu\nu}) + \frac{i\theta_\infty}{32\pi^2} \int \text{Tr}(F \wedge F)$$

The Kapustin—Witten S-duality is exactly:

$$\tau_\infty = \frac{\theta_\infty}{2\pi} + \frac{4\pi i}{g_\infty^2} \xrightarrow{S} -\frac{1}{\tau_\infty}$$

mapping $G \leftrightarrow {}^L G$. [established]

2.2 p -Adic Places ($v = p$)

At a finite prime p , the adelic Yang-Mills action reduces to the Chern—Simons form on the Bruhat—Tits tree $\mathcal{T}_p$ (Phase 3.1):

$$S_p[A] = \frac{k_p}{4\pi} \int_{\mathcal{T}_p} \text{Tr} \left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A \right)$$

where k_p is the Chern—Simons level (the “ p -adic anyon level”).

The S-duality at a p -adic place acts as:

$$\tau_p = \frac{k_p}{2\pi i} \xrightarrow{S} -\frac{1}{n_{\mathfrak{g}} \tau_p} = \frac{2\pi i}{n_{\mathfrak{g}} k_p}$$

This implies the dual level $\tilde{k}_p$:

$$\tilde{k}_p = \frac{1}{n_{\mathfrak{g}} k_p}$$

For $\widehat{\mathfrak{su}}(2)_k$, the lacing number $n_{\mathfrak{g}} = 1$, and S-duality sends:

$$R(\text{SU}(2))_k \xrightarrow{S} R(\text{SO}(3))_{\tilde{k}}$$

where $\text{SO}(3) \cong \text{SU}(2)/\mathbb{Z}_2$ is the Langlands dual of $\text{SU}(2)$. **This is the statement of anyon condensation:** the $\text{SU}(2)_k$ anyon model condenses via the $\mathbb{Z}_2$ center to the $\text{SO}(3)_{\tilde{k}}$ anyon model. [my conjecture]

2.3 Anyon Condensation and Level-Rank Duality

The S-duality at finite primes is mathematically equivalent to **level-rank duality** in Chern—Simons theory:

$$\widehat{\mathfrak{su}}(2)_k \simeq \widehat{\mathfrak{su}}(k)_2$$

The right-hand side is the $SU(k)$ theory at level 2. The fusion ring $R(SU(k))_2$ is isomorphic to the character ring of the group S_3 (the symmetric group on 3 letters for $k = 2$) or more generally to the Verlinde algebra of the W -algebra.

Physically, S-duality at a p -adic place is **anyon condensation**: the quasiparticles of the $SU(2)_k$ theory condense via the $\mathbb{Z}_2$ center to the quasiparticles of the dual theory. The condensation is the p -adic avatar of S-duality. [speculative]

3. Adelic S-Duality

3.1 Adelic Coupling

The adelic coupling $\tau_{\mathbb{A}}$ is an **idele** — an element of the idele class group $\mathbb{A}^{\times}/\mathbb{Q}^{\times}$:

$$\tau_{\mathbb{A}} = (\tau_{\infty}, \tau_2, \tau_3, \dots)$$

with $\tau_p = 1$ (the unramified value) at almost all p . The adelic S-duality transformation acts componentwise:

$$S_{\mathbb{A}} : \tau_{\mathbb{A}} \mapsto (S_{\infty}(\tau_{\infty}), S_2(\tau_2), S_3(\tau_3), \dots)$$

where $S_v(\tau_v) = -1/n_{\mathfrak{g}}\tau_v$ at each place v . [my conjecture]

3.2 Global Langlands from Adelic S-Duality

Theorem 1 (Adelic S-Duality = Global Langlands). The adelic S-duality transformation $S_{\mathbb{A}}$ on adelic Yang-Mills theory is equivalent to the global Langlands correspondence for the Langlands dual group ${}^L G$ over the adèle ring $\mathbb{A}$.

Reasoning. At each place v , the local S-duality S_v is equivalent to the local Langlands correspondence $G(\mathbb{Q}_v) \leftrightarrow {}^L G(\mathbb{Q}_v)$ (Phase 1 and Kapustin—Witten for $v = \infty$). Since both S-duality and the Langlands correspondence factorize over places (as restricted tensor products), the equivalence extends globally:

$$S_{\mathbb{A}} \leftrightarrow \text{Global Langlands for } {}^L G \text{ over } \mathbb{A}$$

The automorphic representations of $G(\mathbb{A})$ classified by the global Langlands correspondence are exactly the S-duality-invariant states in the adelic Hilbert space. [my conjecture]

3.3 Hecke Operator Invariance

The Hecke operators $\mathbf{T}_p$ (Phase 2.1) satisfy an S-duality covariance relation:

$$S_{\mathbb{A}} \circ \mathbf{T}_p \circ S_{\mathbb{A}}^{-1} = \tilde{\mathbf{T}}_p$$

where $\tilde{\mathbf{T}}_p$ are the Hecke operators for the Langlands dual group ${}^L G$. This means:

$$\langle \pi | \mathbf{T}_p | \pi \rangle = \langle {}^L \pi | \tilde{\mathbf{T}}_p | {}^L \pi \rangle$$

The Hecke eigenvalues (Frobenius parameters α_p) are S-duality invariant — a direct physical consequence of the Langlands correspondence. [my conjecture]

4. Adelic Montonen—Olive Duality

4.1 Electric-Magnetic Duality at All Places

The original Montonen—Olive conjecture (1977) proposed that Yang-Mills theory at coupling e is dual to the same theory at coupling $1/e$, with electric charges mapped to magnetic charges. In $N = 4$ SYM, this becomes the full $\mathrm{SL}(2, \mathbb{Z})$ S-duality group at the ∞ -place.

In the adelic framework, electric-magnetic duality extends to every p -adic place:

$$e_v \longleftrightarrow \frac{1}{e_v}$$

where e_v is the coupling at place v . The electric charge at place v is the anyon label j_v ; the magnetic charge is the dual label $\tilde{j}_v$ in the S-dual fusion ring.

Conjecture 2 (Adelic Montonen—Olive). The adelic Yang-Mills path integral is invariant under componentwise S-duality:

$$Z_{\mathrm{ALP}}[\tau_{\mathbb{A}}] = Z_{\mathrm{ALP}}[S_{\mathbb{A}}(\tau_{\mathbb{A}})]$$

This is the adelic generalization of the $N = 4$ SYM S-duality. [my conjecture]

4.2 Physical Spectrum

The S-duality-invariant physical spectrum consists of:

1. **∞ -place:** BPS states of $N = 4$ SYM (dyons with electric and magnetic charge).
2. **p -adic places:** Anyon types that are invariant under the condensation map — these are the **unramified anyons**.
3. **Global object:** The adelic BPS spectrum — the set of automorphic representations satisfying $S_{\mathbb{A}}|\pi\rangle = |\pi\rangle$. [speculative]

5. Verification Against Kapustin—Witten (2006)

5.1 Reduction at the ∞ -Place

The adelic S-duality framework must reproduce the Kapustin—Witten theorem at the ∞ -place. This requires:

1. The ∞ -component of the adelic Yang-Mills theory is $N = 4$ SYM on $\mathbb{R}^4$ (established in Phase 3.1).
2. The surface operator insertion (a ramification at the ∞ -place) corresponds to the choice of a Riemann surface $\Sigma \subset \mathbb{R}^4$.
3. The S-duality S_∞ with the surface operator is equivalent to the geometric Langlands correspondence.

Verification. The adelic coupling $\tau_\infty = \theta/2\pi + 4\pi i/g^2$ maps to $-1/\tau_\infty$ under S_∞ . The Langlands dual group ${}^L G$ for $G = \mathrm{SU}(2)$ is ${}^L G = \mathrm{SO}(3) = \mathrm{SU}(2)/\mathbb{Z}_2$, matching the Kapustin—Witten identification. The Kapustin—Witten result is the ∞ -place specialization of the adelic S-duality framework. ■ [established via Kapustin—Witten]

5.2 Extension to $p = 2$ (Dyadic Place)

At $p = 2$, Phase 1.1 established that $R(\mathrm{SU}(2))_k \cong R(\mathbb{Z}_2^\times)$ for $k + 2 = 2^m$. The S-duality S_2 maps this to:

$$R(\mathrm{SO}(3))_{\tilde{k}} \cong R(\mathbb{Z}_2^\times)^{\text{condensed}}$$

The condensed character ring corresponds to the invariant subspace under $\mathbb{Z}_2$ -twist — exactly the Langlands dual data at $p = 2$. [my conjecture]

5.3 Extension to $p = 3$

At $p = 3$ (Phase 1.2), the ternary level condition $k = 2 \cdot 3^{m-1} - 1$ and the S-duality S_3 produce the $\mathrm{SO}(3)$ theory at the dual level. The anyon condensation removes the $\mathbb{Z}_2$ center, matching the Langlands dual group structure.

6. Falsifiable Predictions

1. **[P1] Adelic S-duality invariance.** The adelic Yang-Mills path integral satisfies $Z[\tau_\mathbb{A}] = Z[S_\mathbb{A}(\tau_\mathbb{A})]$ componentwise. [my conjecture]
 - **Falsification:** Compute the path integral numerically at small p and compare the S-duality partners.
 - **[CHECK: 2028-12]** Phase 3.3 lattice simulations.
2. **[P2] Anyon condensation = S-duality at $p = 2, 3$.** The $\mathrm{SU}(2)_k \rightarrow \mathrm{SO}(3)_{\tilde{k}}$ condensation is exactly the p -adic S-duality transformation, with the fusion ring isomorphism $R(\mathrm{SU}(2))_k \rightarrow R(\mathrm{SO}(3))_{\tilde{k}}$ matching the local Langlands transfer. [my conjecture]
 - **Falsification:** Compute the condensation map from Verlinde algebra and compare to the S-duality prediction.
 - **[CHECK: 2027-12]** Direct computation for $k = 2, 4, 8$.

3. [P3] **Hecke eigenvalue invariance.** Under S-duality, the Hecke eigenvalues $\alpha_p(j)$ are invariant: $\alpha_p(j) = \tilde{\alpha}_p(\tilde{j})$ where $\tilde{j}$ is the S-dual anyon label. [my conjecture]
 - **Falsification:** Find a counterexample Hecke eigenvalue pair.
 - [CHECK: 2028-06] Numerical verification at $p = 2, 3$.
 4. [P4] **Kapustin—Witten at all p .** The geometric Langlands correspondence (Kapustin—Witten 2006) at the ∞ -place is the ∞ -component of a single adelic object that also encodes the local Langlands correspondences at all finite primes via the Silent Parameter. [speculative]
 - **Falsification:** Show that the adelic S-duality framework contradicts a known result about geometric Langlands.
 - [CHECK: 2029-12]
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7. Conclusion

Adelic S-duality extends the Kapustin—Witten correspondence from the Archimedean place to every completion of $\mathbb{Q}$. The componentwise transformation $\tau_v \rightarrow -1/n_{\mathfrak{g}}\tau_v$ at each place v maps the fusion ring $R(G)_k$ to $R({}^L G)_{\tilde{k}}$ at finite primes (anyon condensation) and reproduces $N = 4$ SYM S-duality at ∞ . The global S-duality $S_{\mathbb{A}}$ is the physical realization of the global Langlands correspondence for ${}^L G$ over the adèle ring.

Phase 3 of the ALP program now consists of: - **Phase 3.1:** Adelic Yang-Mills theory — complete - **Phase 3.2:** Adelic S-duality (this paper) — complete - **Phase 3.3:** Numerical adelic lattice simulations — pending

Phase 4 will formulate experimental predictions and measurement protocols based on the adelic gauge theory framework.

Declarations

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References

1. ALP Phase 1—3.1. (2026). Program deliverables, v0.2—v0.9. QNFO Research.
2. Kapustin, A., & Witten, E. (2006). Electric-magnetic duality and the geometric Langlands program. *CMP*, 1(1), 1—236. arXiv:hep-th/0604151.
3. Montonen, C., & Olive, D. (1977). Magnetic monopoles as gauge particles? *Physics Letters B*, 72(1), 117—120.
4. Witten, E. (1989). Quantum field theory and the Jones polynomial. *CMP*, 121(3), 351—399.

5. Verlinde, E. (1988). Fusion rules and modular transformations in 2D CFT. *Nuclear Physics B*, 300, 360—376.
6. Kitaev, A. (2003). Fault-tolerant quantum computation by anyons. *Annals of Physics*, 303(1), 2—30.
7. Bais, F. A., & Slingerland, J. K. (2009). Condensate-induced transitions between topologically ordered phases. *Physical Review B*, 79(4), 045316.
8. Langlands, R. P. (1970). Problems in the theory of automorphic forms. *Lecture Notes in Mathematics*, 170, 18—61.
9. Gelbart, S. (1975). *Automorphic Forms on Adele Groups*. Annals of Mathematics Studies, 83.
10. QNFO Research. (2026). The Langlands Program Is Adelic Physics Without the Physics. Rosetta Note v1.0.